

pulse width modulation how it works



AI Mode

All

Videos

Images

Shopping

Forums

Short videos

More ▾

Tools ▾



AI Overview



Pulse width modulation (PWM) works by rapidly turning a digital signal on and off to control the average voltage and power delivered to a load. It achieves this by varying the "on" time, known as the duty cycle, while keeping the signal's frequency constant. For example, if an LED is pulsed with a 50% duty cycle, it will appear to be half as bright as a continuous signal because it receives power for half of each cycle. [↗](#)

This video explains the basics of pulse width modulation (PWM) and how it works:

Show more ▾



GeeksforGeeks

<https://www.geeksforgeeks.org> › electronics-engineering

Pulse Width Modulation (PWM)

Jul 23, 2025 — **PWM is a modulation method that changes the pulse signal's width** in electrical systems to regulate the average power supplied to a load.



Wikipedia

<https://en.wikipedia.org> › wiki › Pulse-width_modulation

Pulse-width modulation

PWM is useful for controlling the average power or amplitude delivered by an electrical signal. The average value of voltage (and current) fed to the load is ...

History

Principle

PWM sampling theorem

Applications



Basic Electronics Tutorials

<https://www.electronics-tutorials.ws> › blog › pulse-widt...

Pulse Width Modulation Used for Motor Control Applications



<https://learn.sparkfun.com/pulse-width-modulation> > all

Pulse Width Modulation - SparkFun Learn

Pulse Width Modulation (PWM) is a digital signal that varies the time a signal is high, measured by duty cycle, the percentage of time the signal is on.



Control.com

<https://control.com> > Technical Articles

Understanding the Basics of Pulse Width Modulation (PWM)

Mar 23, 2022 — **PWM reduces power by changing the time a device receives full voltage**, using a duty cycle to control the average voltage.

Videos :



7:28

Pulse Width Modulation (PWM) - Electronics Basics 23

YouTube · Simply Electronics

Oct 9, 2016

Pulse Width Modulation (PWM) controls brightness of lights, speed of motors, and efficiency of power supplies by pulsing power on and off.



3:48

Pulse Width Modulation (PWM) Basics

YouTube · WaveShapePlay

Aug 11, 2021

PWM controls power to a load by varying the duty cycle, not amplitude.



13:45

Understanding Pulse Width Modulation

YouTube · Rohde & Schwarz

Feb 26, 2024

Pulsewidth modulation (PWM) controls power delivery by varying the **width** of voltage pulses. Common applications include visual devices, audio, power supplies, and DC motors.

[Feedback](#)

[View all](#) >





Simply put, **modulating the pulse width effectively modulates the average voltage applied to the load.** Higher average voltage results in brighter ...

77 answers · Top answer: Motors are inductors and resist change in current (Lenz's Law). So at a suffici...



BYJU'S

<https://byjus.com> › physics › pulse-width-modulation

How is a Pulse Width Modulation Signal generated?

Pulse width modulation **reduces the average power delivered by an electrical signal** by converting the signal into discrete parts. In the PWM technique, the ...



Same Sky

<https://www.sameskydevices.com> › blog › pulse-width-...

Pulse Width Modulation (PWM): How it Works and Why it's ...

Pulse Width Modulation (PWM) is a control method that reduces average power by rapidly switching a signal on and off, varying the 'on-time' (duty cycle).



Instructables

<https://www.instructables.com> › ... › Microcontrollers

How to Use Pulse-Width Modulation :7 Steps

Pulse-width modulation (PWM) **converts digital to analog signals** by controlling how long a digital signal is held high and how often it switches.

Step 1: Materials That I...

Step 3: The Theory Behind...

Step 4: Operating A Pwm...

People also search for

Pulse width modulation **circuit**



Pulse width modulation **waveform**



Pulse width modulation **PDF**



Pulse width Modulation **block diagram explanation**



Pwm signal pulse width modulation how it works



Pulse width modulation **techniques**



Pulse width Modulation **block diagram**



Pulse Width modulation **Controller**





Central Austin, Austin, TX - [From your IP address](#) - [Update location](#)

[Help](#) [Send feedback](#) [Privacy](#) [Terms](#)